

ORIGINAL ARTICLE

Gut microbiota trajectory in β -thalassemia major children who underwent allogeneic hematopoietic stem cell transplantation

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Abstract

Background: The gut microbiota of patients undergoing allogeneic hematopoietic stem cell transplantation (HSCT) changes, leading to complications such as acute graft-versus-host disease (GVHD). This study aimed to evaluate the human microbiota composition before and after HSCT in β -thalassemia major (β -TM) children.

Method: Twenty-two β -TM children who received allo-HSCT between December 2018 and March 2020 were enrolled. They were followed up for more than 100 days after HSCT, and their gut microbiota information and disease data were recorded at five-time points.

Results: The dominant bacteria were Bacteroidetes and Firmicutes at the phylum level and Lachnospiraceae at the family level before and after HSCT. In the differential analysis, Ruminococcaceae constantly decreased after HSCT. Besides, *Rothia mucilaginosa* was the most abundant 2 months after HSCT compared to before it. Additionally, GVHD patients presented decreased levels of Bacteroidetes compared to those without GVHD. Moreover, Blautia levels significantly decreased in critically ill GVHD patients.

Conclusion: The gut microbiota of the 22 β -TM children showed a clear trend of destruction and reconstruction within 100 days after HSCT. The extra-oral infections and inflammations of *Rothia mucilaginosa*, a Gram-positive bacterium of the normal oropharyngeal tract microbiota, might play an important role in the recovery process of HSCT. Finally, decreased Bacteroidetes levels were associated with GVHD onset.

KEYWORDS

diversity, graft-versus-host disease, juvenile

1 | INTRODUCTION

β -thalassemias major (β -TM) is a heterogeneous autosomal recessive hereditary anemia characterized by inhibited β -globin chain synthesis. The frequency of β -thalassemia varies among regions from <1% to 15%.¹ Approximately 56 000 infants are born with thalassemias each

year.² Although the survival of thalassemia patients has improved, and most patients survive beyond adulthood, the quality of life and ultimate life expectancy remains suboptimal.³ Most thalassemia patients require regular blood transfusion and iron chelation therapy. According to a British study, the state financial expenditure for managing β -TM could be reduced by up to 37% if half of the patients underwent bone marrow transplants, improving the health-related quality of life.⁴

Hematopoietic stem cell transplantation (HSCT) is the only cure available for β -TM. The best HSCT outcome, with overall survival

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(OS) of 80–90%, was reported with human leukocyte antigen (HLA)-matched sibling donors to patients younger than 14 years who did not have significant iron-related complications.^{5,6} However, HSCT might lead to dysbiosis and disruption of intestinal homeostasis due to conditioning regimen, broad-spectrum antibiotics, nutritional changes, intestinal inflammation, and donor cell-derived immune reconstitution. In recent years, increasing evidence has indicated that posttransplant microbiota deviation is closely associated with allo-HSCT outcomes, including infectious complications, graft-versus-host disease (GVHD), disease relapse, and mortality.⁷

Previous studies have demonstrated the role of gut microbiota in metabolic, cardiovascular, and mental diseases.⁸ However, little research on hematological diseases is available, especially in pediatric patients. There are also very limited data tracing the allo-HSCT process. Herein, next-generation sequencing (NGS) was applied to characterize intestinal microbiota compositions in a prospective pilot cohort of allo-HSCT pediatric β -TM recipients.

2 | MATERIALS AND METHODS

2.1 | Ethics statement

The Ethics Committee of the Guangzhou Women and Children's Medical Center approved this study (number: 2019110113014196). Written informed consent was obtained from the guardian of patients before allo-HSCT.

2.2 | Participant recruitment and sample collection

Twenty β -TM children who received allo-HSCT between December 2018 and March 2020 were enrolled in this study. All patients received busulfan (BU) + cyclophosphamide (CY) + anti-thymocyte globulin (ATG) as conditioning regimens. The inclusion criteria were: (1) genetically diagnosed with β -TM⁹; (2) normal cognitive function and communication; (3) under 18 years and without limb deformity; (4) absence of metabolic diseases. Patients were further assigned to the acute GVHD (aGVHD) and no-aGVHD groups if aGVHD occurred within 100 days after HSCT. All children enrolled were followed up for more than 100 days after HSCT. Disease data were recorded at five-time points: pre-HSCT, and 2 weeks, 1 month, 2 months, and 100 days after HSCT.

2.3 | Sample collection, DNA extraction, and sequencing

Fecal samples were collected at the abovementioned time points, preserved in a sterile fecal collection cup, then stored at -80°C . Bacterial genomic DNA was extracted using the E.A.N.A. DNA PowerSoil DNA Kit (Omega, USA). The reverse transcription-polymerase chain reaction (RT-PCR) was performed using a PCR kit (Transgene, CN). The

V4-V5 regions of 16S ribosomal DNA sequences were used for DNA library construction using 515F-907R primers. The qualified amplification libraries were then sequenced (250 bp paired reads) using the Illumina NovaSeq 6000 platform (Illumina, USA).

2.4 | Bioinformatics analysis

The sequenced raw data were assigned to each sample using the barcode identifier. Low-quality sequences were removed using Trimmomatic (v. 0.39).¹⁰ Then, the available paired-end reads were connected based on the overlap using FLASH (v. 1.2.11).¹¹ Subsequently, these assembled tags were clustered into operational taxonomic units (OTUs) under 97% sequence similarity using the Usearch software.¹² Representative OTU sequences were annotated with microbial taxonomy against the RDP database using the Qiime package.^{13,14} The bacterial function was predicted using PICRUST.¹⁵ The bacterial abundance and diversity of each sample were calculated using Mothur software,¹⁶ and the composition trend was obtained by Principal Component Analysis (PCA).

2.5 | Statistical analysis

All statistical analyses were conducted in R (v. 3.6.0). Differential microbial abundance between different groups or time points ($p < .05$) was assessed by the Wilcoxon rank-sum test using the “stats” R package. Nonparametric multivariate analysis of variance (PERMANOVA) (“vegan” R package) was used to determine the influence of each factor on the microbiota composition. The Shannon Index¹⁷ was calculated using Mothur (v. 1.39.5). KEGG pathways were predicted using Picrust. The weighted UniFrac distance was analyzed using QIIME (v. 1.9.1). The confidence level was $\alpha = .05$. Data visualization was performed using the “ggplot2” R package.

3 | RESULTS

3.1 | Clinical features

Twenty-two β -TM children (mean age: 6.93 ± 3.17 years) who underwent allo-HSCT in 2019 were enrolled, and three have died until now. Clinical and transplant characteristics are summarized in Table 1. All patients accepted stem cells from peripheral blood, and the 16S rRNA gene sequencing was performed for 106 stool samples before and after HSCT.

3.2 | Gut microbiota diversity

The Shannon index was applied to estimate the alpha diversity.¹⁷ A clear gradual increasing trend was detected after the sharp flop (Figure S1). Two weeks after transplantation, the Shannon index decreased,

TABLE 1 Clinical characteristics of β -TM children who underwent allo-HSCT.

Age at transplantation (median [range]) (years)	6.55 (3.25–16.58)
Gender (male/female)	
Male	12
Female	10
Graft source	
Marrow bank	13
Sibling	6
Mother	2
Father	1
Infection after transplantation	
Infected	13
Uninfected	9
Donor HLA-match/relation	
Matched related	7
Mismatched unrelated	5
Matched unrelated	9
Haploidentical	1
Stem cell source	
Peripheral blood	22
Conditioning regimen	
Myeloablative intensity	22
Acute GVHD level	
I	3
II	1
IV	2
Acute GVHD site	
Skin	4
Gut	2
Ferritin levels (average [range]) (ng/ml)	3091.1 (608.4–10286.7)
Steroid use	
Yes	7
No	15

Note: Recipient information is summarized by count (percentage) for categorically coded variables.

Abbreviation: HSCT, hematopoietic stem cell transplantation.

suggesting that the gut microbiota abundance was disrupted, and after 100 days, most patients recovered to a similar pre-transplantation level. The Bray-Curtis dissimilarity indicated that the gut microbiota composition mainly varied 1 month after HSCT and was even more alike 100 days post-HSCT compared to before the HSCT.

The PCA reduces the dimensionality of the original complex data and reveals the simple structure hidden behind the complex data. Herein, the PCA suggested a piecemeal deviation of the gut microbiota structure over time. The beta diversity and PCA showed that gut microbiota was the most similar to pre-HSCT 2 months after HSCT

compared to other time points. In the beta analysis, all time points exhibited statistically different results except for pre-HSCT versus 2-month-post-HSCT.

3.3 | Community composition before HSCT

Before HSCT, Bacteroidetes (29.2%) and Firmicutes (55.3%) were the dominant bacteria at the phylum level, and Clostridia (54.9%) and Bacteroidia (29.2%) were the most abundant classes. Before the transplant regimen, the most abundant taxa at the family level in children was Lachnospiraceae (33.4%), mainly *Blautia* and *Lachnospiraceae* incertae sedis. Regarding the 10 most abundant species, five belonged to the order Eubacteriales.

3.4 | Community composition after HSCT

Furthermore, the bacterial abundance was analyzed at different time points. The results with *p*-value and false discovery rate < 0.05 are listed in Table 2. The gut microbiota composition before HSCT was very different than after HSCT (Figure S2). However, no statistical differences were detected at the phylum and class levels. Meanwhile, Actinomycetales and Micrococcaceae were enriched 2 months after HSCT, and a constant decrease of Ruminococcaceae was detected after HSCT. For most bacteria at the genus level, the abundance fits the decreasing tendency 1 month after HSCT and approximately recovered to the pre-HSCT level within the next month, including *Anaerostipes*, *Blautia*, *Collinsella*, *Faecalibacterium*, *Fusicatenibacter*, and *Ruminococcus* (Table 2). *Rothia* was the only genus significantly enriched 2 months after HSCT, whereas the others presented higher abundance before treatment. Interestingly, *Bacteroides uniformis* was enriched at time four compared to time two, becoming the only bacterium with significantly changed abundance after HSCT. *Rothia mucilaginosa*, *Collinsella aerofaciens*, and *Gemmiger formicilis* showed the most noticeable changes at the species level.

3.5 | Individual conditions of GVHD children

Four of the seven children who developed GVHD (Figure 1) had skin GVHD and one grade IV gut GVHD. The LJZ patient developed gut GVHD on day 36 after infusion and was cured within 100 days. His gut microbiota mainly changed at genus and species levels, showing a significant increase in *Blautia wexlerae* levels. He was diagnosed with lung infection before GVHD and CMV and EBV infection during GVHD. Four skin GVHD patients got the disease between 14 days and 1 month. After the GVHD onset, two got infected, and their dominant bacterial communities converted to Bacteroides. The other two children without infection presented an increase in Actinomyces and Bifidobacterium, respectively. Skin acute GVHD (aGVHD) grading¹⁸ and immunosuppression agents used to treat aGVHD are listed in Table S1.

**TABLE 2** Distinctly different taxes.

Timepoints	Level	Tax	Abundance at the earlier timepoint	Abundance at the later timepoint	Enriched	p-Value	FDR
Time 1 versus Time 4	order	Actinomycetales	0.47 ±1.16	6.93 ± 12.40	Time 4	.004	0.030
Time 1 versus Time 4	family	Micrococcaceae	0.14 ±0.20	4.76 ± 9.12	Time 4	.002	0.019
Time 1 versus Time 2	family	Ruminococcaceae	15.39 ±12.21	10.88 ± 21.47	Time 1	.002	0.022
Time 1 versus Time 3	family	Ruminococcaceae	15.39 ±12.21	6.20 ± 10.34	Time 1	.000	0.006
Time 1 versus Time 4	family	Ruminococcaceae	15.39 ±12.21	4.34 ± 4.71	Time 1	.000	0.005
Time 1 versus Time 5	family	Ruminococcaceae	15.39 ±12.21	4.22 ± 5.22	Time 1	.000	0.001
Time 1 versus Time 3	genus	<i>Alistipes</i>	0.56 ±0.93	0.12 ± 0.16	Time 1	.014	0.036
Time 1 vs. Time 2	genus	<i>Anaerostipes</i>	0.96 ±1.54	0.12 ± 0.12	Time 1	.001	0.012
Time 1 vs. Time 3	genus	<i>Anaerostipes</i>	0.96 ±1.54	0.21 ± 0.43	Time 1	.001	0.007
Time 1 vs. Time 2	genus	<i>Blautia</i>	12.78 ±9.84	3.45 ± 3.40	Time 1	.000	0.004
Time 1 vs. Time 3	genus	<i>Blautia</i>	12.78 ±9.84	5.91 ± 8.62	Time 1	.002	0.007
Time 1 vs. Time 2	genus	<i>Collinsella</i>	3.23 ±5.84	0.28 ± 0.57	Time 1	.004	0.023
Time 1 vs. Time 3	genus	<i>Collinsella</i>	3.23 ±5.84	0.25 ± 0.52	Time 1	.001	0.007
Time 1 vs. Time 3	genus	<i>Dorea</i>	1.40 ±1.87	0.15 ± 0.18	Time 1	.006	0.021
Time 1 vs. Time 2	genus	<i>Faecalibacterium</i>	7.85 ±11.58	1.19 ± 1.57	Time 1	.002	0.012
Time 1 vs. Time 3	genus	<i>Faecalibacterium</i>	7.85 ±11.58	2.83 ± 5.35	Time 1	.015	0.036
Time 1 vs. Time 5	genus	<i>Faecalibacterium</i>	7.85 ±11.58	1.04 ± 1.04	Time 1	.001	0.037
Time 1 vs. Time 2	genus	<i>Fusicatenibacter</i>	1.12 ±1.56	0.11 ± 0.11	Time 1	.008	0.030
Time 1 vs. Time 3	genus	<i>Fusicatenibacter</i>	1.12 ±1.56	0.82 ± 3.42	Time 1	.012	0.035
Time 1 vs. Time 2	genus	<i>Gemmiger</i>	4.01 ±7.40	0.55 ± 0.86	Time 1	.008	0.030
Time 1 vs. Time 3	genus	<i>Gemmiger</i>	4.01 ±7.40	0.33 ± 0.52	Time 1	.001	0.007
Time 1 vs. Time 4	genus	<i>Gemmiger</i>	4.01 ±7.40	0.29 ± 0.51	Time 1	.000	0.012
Time 1 vs. Time 3	genus	<i>Roseburia</i>	1.33 ±3.03	0.47 ± 1.60	Time 1	.018	0.040
Time 1 vs. Time 4	genus	<i>Rothia</i>	0.10 ±0.13	4.73 ± 9.08	Time 4	.001	0.012
Time 1 vs. Time 2	genus	<i>Ruminococcus</i>	1.60 ±2.82	0.30 ± 0.45	Time 1	.012	0.041
Time 1 vs. Time 3	genus	<i>Ruminococcus</i>	1.60 ±2.82	0.22 ± 0.44	Time 1	.002	0.007
Time 1 vs. Time 3	genus	<i>Ruminococcus2</i>	2.32 ±2.38	1.21 ± 2.44	Time 1	.002	0.007
Time 1 vs. Time 2	species	<i>Anaerostipes_hadrus</i>	0.93 ±1.55	0.10 ± 0.12	Time 1	.008	0.033
Time 1 vs. Time 3	species	<i>Anaerostipes_hadrus</i>	0.93 ±1.55	0.19 ± 0.44	Time 1	.012	0.041
Time 2 vs. Time 4	species	<i>Bacteroides_uniformis</i>	0.63 ±1.14	1.77 ± 2.33	Time 4	.002	0.047
Time 1 vs. Time 2	species	<i>Blautia_faecis</i>	0.68 ±0.70	0.13 ± 0.16	Time 1	.000	0.004
Time 1 vs. Time 3	species	<i>Blautia_faecis</i>	0.68 ±0.70	0.10 ± 0.12	Time 1	.000	0.002
Time 1 vs. Time 2	species	<i>Blautia_wexlerae</i>	9.31 ±7.49	1.71 ± 1.91	Time 1	.000	0.001
Time 1 vs. Time 3	species	<i>Blautia_wexlerae</i>	9.31 ±7.49	2.28 ± 4.88	Time 1	.000	0.002
Time 1 vs. Time 4	species	<i>Blautia_wexlerae</i>	9.31 ±7.49	5.83 ± 16.05	Time 1	.003	0.026
Time 1 vs. Time 3	species	<i>Collinsella_aerofaciens</i>	2.19 ±4.00	0.12 ± 0.28	Time 1	.001	0.007
Time 1 vs. Time 2	species	<i>Faecalibacterium_prausnitzii</i>	7.85 ±11.58	1.19 ± 1.57	Time 1	.002	0.014
Time 1 vs. Time 3	species	<i>Faecalibacterium_prausnitzii</i>	7.85 ±11.58	2.83 ± 5.35	Time 1	.015	0.045
Time 1 vs. Time 5	species	<i>Faecalibacterium_prausnitzii</i>	7.85 ±11.58	1.04 ± 1.04	Time 1	.001	0.040

(Continues)

TABLE 2 (Continued)

Timepoints	Level	Tax	Abundance at the earlier timepoint	Abundance at the later timepoint	Enriched	p-Value	FDR
Time 1 vs. Time 2	species	<i>Fusicatenibacter_saccharivorans</i>	1.12 ± 1.56	0.11 ± 0.11	Time 1	.008	0.033
Time 1 vs. Time 3	species	<i>Fusicatenibacter_saccharivorans</i>	1.12 ± 1.56	0.82 ± 3.42	Time 1	.012	0.041
Time 1 vs. Time 2	species	<i>Gemmiger_formicilis</i>	4.01 ± 7.40	0.55 ± 0.86	Time 1	.008	0.033
Time 1 vs. Time 3	species	<i>Gemmiger_formicilis</i>	4.01 ± 7.40	0.33 ± 0.52	Time 1	.001	0.007
Time 1 vs. Time 4	species	<i>Gemmiger_formicilis</i>	4.01 ± 7.40	0.29 ± 0.51	Time 1	.000	0.012
Time 1 vs. Time 3	species	<i>Roseburia_inulinivorans</i>	1.23 ± 2.98	0.47 ± 1.60	Time 1	.016	0.045
Time 1 vs. Time 4	species	<i>Rothia_mucilaginosa</i>	0.10 ± 0.13	4.72 ± 9.05	Time 4	.001	0.012
Time 1 vs. Time 3	species	<i>Ruminococcus_bromii</i>	1.47 ± 2.76	0.20 ± 0.42	Time 1	.003	0.013
Time 1 vs. Time 3	species	<i>Ruminococcus_faecis</i>	2.22 ± 2.34	1.11 ± 2.35	Time 1	.001	0.007

Note: FDR means false discovery rate.

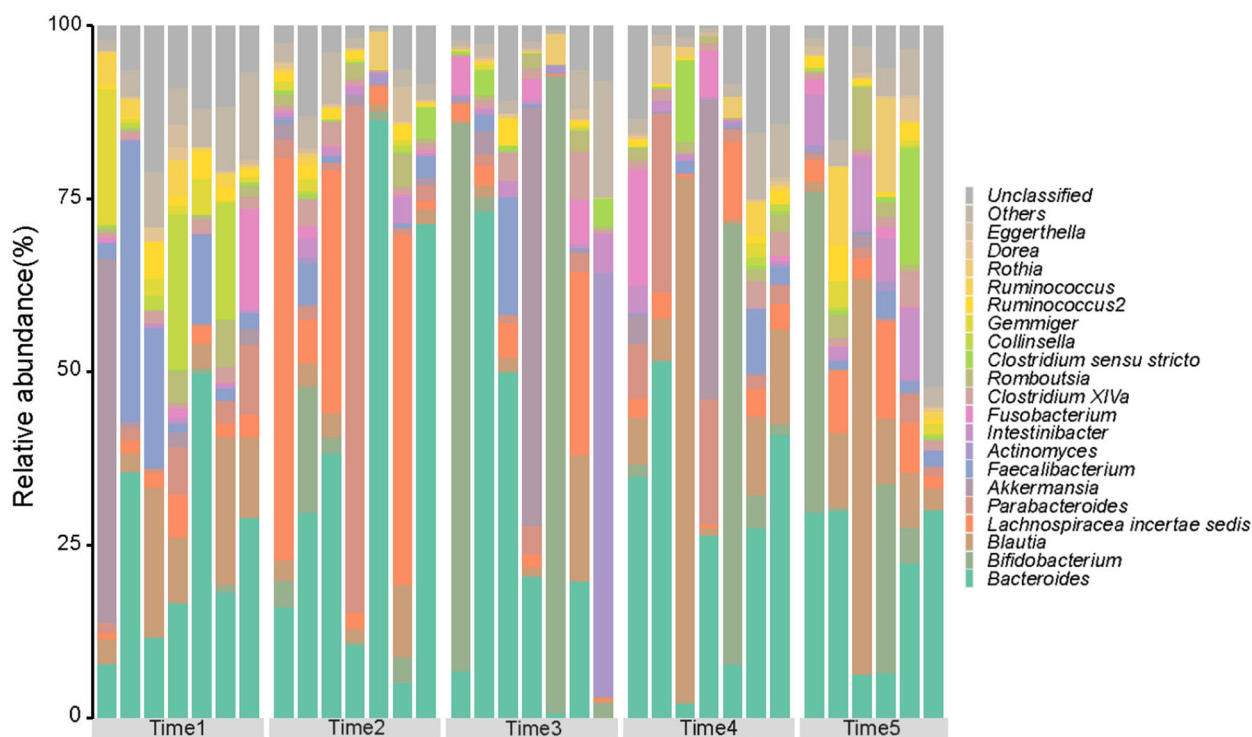


FIGURE 1 Genus abundance of GVHD patients. Gut microbiota composition and abundance of children with β -thalassemia major undergoing allo-HSCT at genus level. Stool samples were collected before the transplantation regimen (time 1), +2 weeks (time 2), +1 month (time 3), +2 months (time 4) and 100 days after transplantation (time 5).

3.6 | Differences between the no-GVHD and the GVHD groups

None of the taxa significantly correlated with GvHD (Figure S3), which could stem from inadequate statistical power. At the phylum level, GVHD patients showed decreased Bacteroidetes after onset com-

pared to non-GVHD patients. *Bifidobacterium* was the most discrepant bacterium before transplantation. Two weeks after HSCT, *Prevotella*, *Flavonifractor*, *Eggerthella*, *Akkermansia*, and *Clostridium sensu stricto* showed clear variance. Most GVHD patients developed the disease 1 month after HSCT, but the gut microbiota at this time showed little difference. Two months after HSCT, several bacteria differed in the

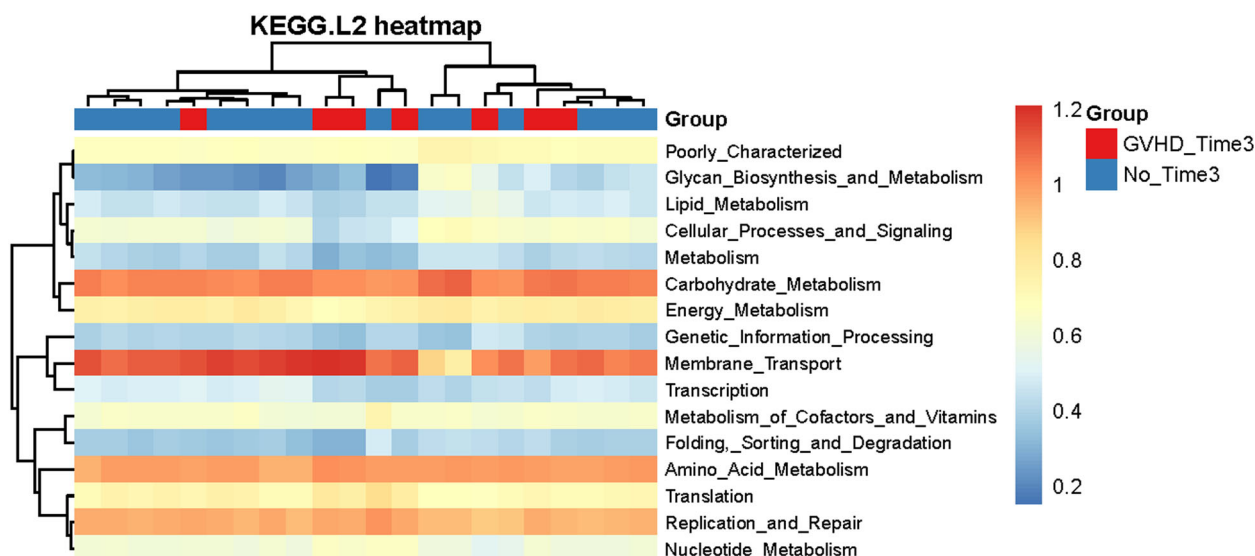


FIGURE 2 KEGG analysis 1 month after HSCT. The color intensity indicates the strength of each patient's involvement in a specific pathway or biological process. Darker colors indicate higher expression levels or greater activity.

two groups, including *Robinsoniella*, *Parabacteroides*, and *Lachnospiraceae* incertae sedis. The genus *Collinsella* and *Terrisporobacter* varied 100 days after HSCT. The KEGG analysis (Figure 2) was also conducted between no-GVHD and GVHD groups to link genomic information with higher-order functional information. No statistical difference was found between no-GVHD and GVHD groups. Within the GVHD group, several KEGG pathways were enriched right after HSCT compared to before HSCT, including DNA replication proteins, cysteine and methionine metabolism, mismatch repair, nitrogen metabolism, DNA replication, and pantothenate and CoA biosynthesis.

4 | DISCUSSION

Herein, the gut microbiota trajectory showed an evident disruption and reconstruction trend. The Shannon index decreased 2 weeks after transplantation and gradually recovered in the next two and a half months, similar to another study on thalassemia children. In this study, the Shannon index was low and similar between patients before transplantation but significantly lower on days 15 and 30 after transplantation.¹⁹ Another study found that HSCT resulted in a profound disturbance of the gut ecosystem, measured by the loss of about 30% of the pre-HSCT alpha diversity. Additionally, the mean value of weighted UniFrac distances among pre-HSCT samples was significantly lower than post-HSCT samples, demonstrating the increased inter-subject gut microbiota diversity after transplantation. On the other hand, samples obtained more than 30 days after HSCT presented a similar degree of intersubject diversity to pre-HSCT samples, indicating a progressive ecosystem recovery after HSCT.²⁰

Furthermore, at the phylum level, Bacteroidetes and Firmicutes were the most abundant taxa in children in the whole transplant regimen process without much abundance changes. Meanwhile, Turkish scientists reported that, at the phylum level, approximately all reads

were Ascomycota, accounting for 98.4%, 96.9%, 99.1%, and 83.4% before the transplant day and on days 15, 30, and 90, respectively. The children's most abundant taxa at the family level were Saccharomycetaceae (65.2%), followed by Mucoromycota uc and Mucoraceae. Besides, Basidiomycota comprised 15.6% 3 months after HSCT, with *Malassezia* representing 13.8% at the genus level.¹⁹ On the other hand, several HSCT studies were consistent with the current study, suggesting that the gut microbiota of HSCT patients and healthy control subjects mainly comprised Bacteroidetes and Firmicutes.^{20–22} The Lachnospiraceae family was the most abundant in the present study, and the 20 most abundant species mainly belonged to this clade. Previous studies have shown that Lachnospiraceae had greater relative abundance in allo-HSCT recipients who were alive, and their loss was associated with an increased risk for *Clostridium difficile* infections.^{23–25}

The differential analysis showed a constant decrease in Ruminococcaceae after HSCT. Similarly, a single-center study that recruited 37 children observed a reduction in abundances immediately after HSCT from 9.6% before HSCT to 1.7% on average in 5 weeks. Patients with high abundances of obligate anaerobes, such as Ruminococcaceae, developed no or mild aGVHD and exhibited low mortality, showing rapid reconstitution of natural killer (NK) and B cells.²⁶ *Rothia mucilaginosa* showed the clearest enrichment at time four compared to time one, indicating intra-oral translocation. In previous studies, the abundance of Gram-positive oral bacteria, such as *Rothia mucilaginosa*, has been correlated with the development of severe acute GVHD.²⁷

Moreover, this study did not detect any taxon significantly related to GVHD, consistent with the findings with an adult cohort, in which the recipient's pre-conditioning microbiota of the allo-HSCT and GVHD were not significantly correlated.²⁸ After onset, the GVHD patients enrolled here presented a decreased level of Bacteroidetes compared to non-GVHD patients. This finding was consistent with Eriguchi et al., who found that GVHD was associated with lower levels of Firmicutes and Bacteroidetes, which are rarely pathogenic and make several

essential contributions to human physiology and health.²⁹ Sofi et al. also reported that increased *Bacteroides* in the intestinal microbiota of recipients were associated with reduced GVHD in mice after fecal microbial transplantation. Oral gavage administration of *Bacteroides fragilis* increased gut microbiota diversity and beneficial commensal bacteria. The underlying mechanisms might be related to the ability to increase short-chain fatty acids, IL-22, and regulatory T cells, improving the tight junction integrity of the gut and reducing inflammatory cytokine production of pathogenic T cells.³⁰ The GVHD patient death case in the present study presented a significant decrease in *Blautia*, consistent with a multicenter study that found lower *Blautia* and increased *Fusobacterium* abundance among children developing GVHD.³¹ Another study based on two independent cohorts found that the increased *Blautia* was associated with reduced GVHD-related mortality, indicating the potential of medical tactics to prevent harm to the gut microbiome. Specifically, preserving necessary anaerobic microorganisms, such as *Blautia*, should be further explored to enhance the therapeutic outcomes of allo-bone marrow transplantation (BMT) recipients.³²

However, the current study has some limitations, including the relatively small sample size and single-center study. Additionally, the pilot nature of this study hindered the ability to make a definitive conclusion about intestinal microbiota compositions of allo-HSCT pediatric recipients with β -TM. Moreover, the effects of diet on gut microbiota composition were not evaluated. Patients' diets were identical, typically consisting of aseptic diets during hospitalization and a relatively simple diet of rice, pork, and vegetables after discharge. The reason was that the patients' families believed it to be the safest diet. Although clinical nutritionists have intervened, the families were concerned that certain foods would cause diarrhea and hesitated to accept a more diverse diet. Despite these limitations, this study addressed an important topic and filled the gap in tracing allo-HSCT microbiota changes among pediatric patients. The gut microbiome plays a fundamental role in disease development, and these results might help support a rational approach to fecal material transplantation during HSCT.

CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

All data are available in our manuscript.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Luo H, Liu T, Qu Y, et al. Gut microbiota trajectory in β -thalassemia major children who underwent allogeneic hematopoietic stem cell transplantation. *Transpl Infect Dis*. 2023;25:e14111. <https://doi.org/10.1111/tid.14111>